BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, BANNU

SECONDARY SCHOOL (9TH CLASS) EXAMINATION, 2022

MATHEMATICS

Past Paper 2022 • Paper Code: BN-IX-XXII-I

Retyped faithfully from the original board paper — same questions, no changes.

SECTION – A (Version-C)

Time Allowed: 20 Min | **Maximum Marks:** 15 | **Paper Code:** BN-IX-XXII-I

Note: There are THREE sections in this paper i.e. A, B & C. Attempt section A and return it to the superintendent within the given time. You have four choices for each objective type question as A, B, C and D. The choice which you think is correct fill that circle in front of the question number. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero mark in that question.

Q1. Age of x years old boy after 4 years shall be _____ years.

(A) 4
(B) 4x
(C) 4 + x
(D) 4^2

Q2. a < b, b < c ⇒ a < c shows _____ property

(A) Transversal
(B) Transpose
(C) Transitive
(D) Trichotomy

Q3. The line x = a is parallel to _____.

(A) x-axis
(B) y-axis
(C) Both A & B
(D) Neither A nor B

Q4. The point having its coordinates (0,0) is called:

(A) Ordinate
(B) Abscissa
(C) Origin
(D) Critical point

Q5. Any non zero number raised to the zero power equals _____.

(A) -1
(B) 0
(C) 1
(D) Not defined

Q6. L.C.M of $x^2 - x + 1$ and $x^3 + 1$ is _____.

(A) $x^2 - x + 1$ (B) $x^3 + 1$

(C) $(x^2 - x + 1)(x - 1)$ (D) $(x + 1)^3$

Q7. y coordinates of (y, -x) is _____.

(A) y (B) -y

(C) x (D) -x

Q8. The coordinates of midpoint of the segment with end points (2,4) and (0,0) are _____.

(A) (2,4) (B) (1,2)

(C) (0,0) (D) (-2,-4)

Q9. i - (-i) = _____.

(A) -1 (B) 1

(C) i^2 (D) 1

Q10. Base of common logarithm is _____.

(A) 1 (B) 10

(C) π (D) e

Q11. $(a+b)^2 - (a-b)^2$ = _____.

(A) 4ab (B) 0

(C) $2(a^2+b^2)$ (D) 2(a+b)

Q12. $\sqrt{15}/3$ = _____.

(A) $\sqrt{5}$ (B) $\sqrt{5}/3$

(C) $5/\sqrt{3}$ (D) $\sqrt{5}/\sqrt{3}$

Q13. Factors of 12 used in the factorization of $a^2 - 7a + 12$ are _____.

(A) 2,6 (B) 1,2

(C) 3,4 (D) [illegible in source scan]

Q14. H.C.F of (x-3)(x-1) and (x-2)(x-1) is _____.

(A) (x-1) (B) (x-2)(x-3)

(C) (x-2) (D) (x-1)(x-2)

Q15. For the matrices M and N, $(MN)^t$ = _____.

(A) $M^t + N^t$ (B) $M^t - N^t$

(C) M^tN^t (D) N^tM^t

SECTION – B

Time Allowed: 2:40 Hrs | **Maximum Marks:** 36 | **Paper Code:** BN-IX-XXII-I

Note: Attempt section B & C accordingly.

Q2. (Marks 36) Q.No 2. Attempt any NINE parts of the following. All parts carry equal marks.

i. Evaluate $(2x - 3) / (x^2 - x + 1)$ for $x = 2$.

ii. Factorize $2a^2 - 7a + 3$

iii. Find the value of k when $x^3 + 9x^2 - (3k - 4)x + 2$ has the factor $(x - 1)$

iv. If product of two polynomials is $x^4 + 3x^3 - 6x^2 - 2x - 28$ and their HCF is $x - 2$, find their LCM.

v. Simplify $3 / (t - 2) - t / (t^2 - 4)$

vi. Using factorization method, find the square root of $x^2y^2 + 8xy + 16$

vii. Find the values of a, b, c and d when $[[a, 2c],[2c + d, 2a - b]] = [[-1, 4],[8, 0]]$

viii. Simplify $\sqrt{a^6} \cdot \sqrt[3]{a^4}$

ix. Find a when $\log_2 6 + \log_2 7 = \log_2 a$

x. Simplify $x / (x + 2) + (x + 3) / (x - 3)$

xi. Find the solution set of $|3x - 5| = 4$

xii. Evaluate the distance between the two points (3,7) and (0,5)

SECTION – C

Maximum Marks: 24 | **Paper Code:** BN-IX-XXII-I

Note: Attempt any THREE of the following. All questions carry equal marks.

Q3. Q.No:3. Prove that the triangle formed by vertices P(1,-2), Q(5,1) and R(2,5) is a right angled triangle.

Q4. Q.No:4. Prove that any point on the bisector of an angle is equidistant from its arms.

Q5. Q.No:5. The sum of the lengths of any two sides of a triangle is greater than the length of the third side.

Q6. Q.No:6. Draw any triangle XYZ with $m\angle Y = 90°$. Construct its medians and show that they are concurrent.